

East West University

Department of Computer Science and Engineering

Project Title : Motor-Imagery EEG Classification Using Graph Neural Networks with Functional Channel Connectivity

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(EEG-GraphNet)

Introduction

Motor imagery (MI) is the process of imagining a body movement without physically performing it. Electroencephalography (EEG) records brain activity and can be used to identify these imagined movements. However, EEG classification is challenging because signals are noisy, vary between participants, and contain complex relationships between different brain regions.

Recent studies have applied Graph Neural Networks (GNNs) to improve EEG classification. In these models, EEG channels are represented as graph nodes, while edges describe relationships between channels. This approach allows the model to learn spatial and functional information from brain signals.

This review examines five studies published between 2022 and 2025. It focuses on their datasets, methods, graph construction, performance, strengths, limitations, and relevance to the proposed EEG-GraphNet model.

Dataset overview

Participants	EEG Channels	Classes	Sampling rate
9	22	4	250 Hz

The dataset contains nine participants (A01–A09), 22 EEG channels, three additional EOG channels, and four motor-imagery classes. Each participant has two recording sessions, with 288 trials per session. The signals are sampled at 250 Hz and have undergone band-pass and notch filtering.

Related Work Papers (2022-2025)

Paper 1 — Feature learning framework based on EEG graph self-attention networks for motor imagery BCI systems
2023 · Journal of Neuroscience Methods
DOI: 10.1016/j.jneumeth.2023.109969

Paper 2 — Novel channel selection model based on graph convolutional network for motor imagery https://pmc.ncbi.nlm.nih.gov/articles/PMC10542066/?
doi: 10.1007/s11571-022-09892-1

Paper 3 — A brain topography graph embedded convolutional neural network for EEG-based motor imagery classification
https://www.sciencedirect.com/science/article/abs/pii/S1746809424004592?via%3Dihub
DOI: 10.1016/j.bspc.2024.106401

Paper 4 — Graph Convolutional Neural Network with Multi-Scale Attention Mechanism for EEG-Based Motion Imagery Classification
https://www.worldscientific.com/doi/10.1142/S0218001423540204

Paper 5 — Graph vertex and spectral features for EEG-based motor imagery classification
2025 · Computers in Biology and Medicine
DOI: 10.1016/j.compbiomed.2025.109944

Comparative Review of Related Work

Table 1. Comparison of five studies related to graph-based motor-imagery EEG classification

Study: title, authors and year	**Paper 1** Sun et al. (2023)	**Paper 2** Liang et al. (2023; online 2022)	**Paper 3** Shi et al. (2024)	**Paper 4** Zhu et al. (2023)	**Paper 5** Abdelaty et al. (2025)

Dataset and application area	Two MI-EEG datasets	Three MI-EEG datasets	BCI IV-2a, HGD	MI-EEG (dataset unconfirmed)	BCI IV-2a, Dataset 1
Method/model and GNN type	Graph Self-Attention Network	GCN	GCN + CNN + TCN	GCN + Multi-Scale Attention	Graph features + ML (not GNN)
Graph construction and attention type	Spatial channel connections	Pearson correlation	Mutual information	Phase Lag Index (PLI)	Weighted directed graph
Reported metrics (Accuracy)	Not verified	79.76%–89.14%	80.46% (IV-2a)	93.22%	81.75% (IV-2a)
Strengths	Learns useful EEG features	Selects important channels	Learns spatial and temporal features	Captures functional connections	Uses interpretable graph features
Limitations and research gap	Limited graph comparison	Focuses on channel selection	Complex model	Dataset comparison unclear	Relies on handcrafted features
Relation to our work	We will compare different graph structures	We will test channel efficiency	Complex model	We will test connectivity using consistent evaluation	We will compare handcrafted and learned features

Discussion of Related Studies

Paper_1 — Sun et al. (2023) introduced EGSAN to learn useful features from EEG graphs. Their approach used power spectral density features and self-attention to capture information from EEG channels. The study demonstrated the value of graph-based feature learning, particularly when labeled data are limited.

Paper_2 — Liang et al. (2023) developed a GCN-based method for selecting important EEG channels. The authors used Pearson correlation to represent relationships between electrodes. Their results showed that removing unnecessary channels can improve classification accuracy. This is relevant to our project because channel efficiency may help reduce the complexity of EEG classification.

Paper_3 — Shi et al. (2024) proposed a hybrid model combining GCN, CNN and TCN. Mutual information was used to construct the EEG graph. The model achieved 80.46% accuracy on BCI IV-2a, showing that graph-based spatial information can support motor-imagery classification.

Paper_4 — Zhu et al. (2023) developed a GCN with multi-scale attention. They used Phase Lag Index to represent functional connectivity between EEG channels and achieved 93.22% accuracy under tenfold cross-validation. Their study highlights the potential benefit of combining functional connectivity with attention mechanisms. However, this result should not be directly compared with BCI IV-2a results until the experimental dataset and protocol are confirmed.

Paper_5 — Abdelaty et al. (2025) combined graph vertex and spectral features with traditional classification approaches. Their method achieved 81.75% accuracy on BCI IV-2a. Unlike a GNN, their method uses graph-based feature engineering followed by conventional classifiers. This provides a useful reference for examining whether learned graph representations offer advantages over handcrafted features.

Overall, these studies demonstrate that graph-based representations can improve the analysis of motor-imagery EEG signals. However, their results also show the importance of graph design, feature selection, model complexity, and evaluation methods.

Research Gap

Existing studies have already demonstrated the effectiveness of GNNs and graph-based features for motor-imagery EEG classification. Therefore, the proposed research does not claim that graph-based EEG classification is a new concept.

Instead, the main research gap is the need for a controlled and reproducible comparison of EEG graph construction methods, feature representations, and generalization performance on BCI Competition IV Dataset 2a.

Five specific areas will be investigated:

1. **Graph construction:** The effect of electrode-distance, correlation-based, and learnable adjacency matrices on classification performance.

2. **Feature representation:** The usefulness of interpretable band-power and log-variance features compared with learned temporal features.
3. **Generalization:** Differences in classification performance across participants and recording sessions.
4. **Channel efficiency:** Whether fewer EEG channels can maintain satisfactory classification performance.
5. **Fair model comparison:** Evaluation of the proposed GNN against CSP-LDA, CSP-SVM and EEGNet using consistent preprocessing and leakage-free data splits.

Relation to the Proposed EEG-GraphNet Model

Our project aims to classify four motor-imagery tasks: left-hand, right-hand, both-feet and tongue imagery. We use BCI Competition IV Dataset 2a, containing EEG recordings from nine participants and 22 EEG channels sampled at 250 Hz.

In our proposed model, each EEG channel will represent a graph node, while graph edges will represent spatial or functional relationships between electrodes. The GNN will use these relationships to learn patterns associated with imagined movements.

The Task 1 exploratory analysis investigates EEG signal distributions, data quality, class balance, correlations, and extracted features. These features include log-variance and relative power in theta, mu/alpha, beta and gamma frequency bands.

Building on this analysis, our subsequent experiments will examine different graph construction methods and evaluate their effects on classification performance. We will compare the proposed model with established machine-learning and deep-learning approaches, using accuracy and other suitable evaluation metrics.

Performance Benchmarks for Future Comparison

Table 2. Published results most relevant to the proposed evaluation

Study	Dataset	Reported accuracy	Comparability
P1. Sun et al. (2023)	Two MI datasets	Not verified	Requires full results and protocol
P2. Liang et al. (2023)	Three MI datasets	79.76%, 89.14%, 87.96%	Different dataset setups

P3. Shi et al. (2024)	BCI IV-2a	80.46%	Relevant reference
P4. Zhu et al. (2023)	MI-EEG; dataset not confirmed	93.22%	Protocol and dataset require verification
P5. Abdelaty et al. (2025)	BCI IV-2a	81.75%	Relevant reference

The results of **Shi et al. (2024)** and **Abdelaty et al. (2025)** are particularly relevant because both include **BCI IV-2a**. Nevertheless, a fair numerical comparison requires checking whether the same subjects, trials, train-test splits and evaluation procedures were used.

Our final evaluation should report classification accuracy, macro-F1 score, Cohen's kappa, confusion matrices and subject-wise performance where appropriate.

Conclusion

The reviewed studies demonstrate the potential of graph-based methods for motor-imagery EEG classification. They show that electrode relationships, connectivity measures, feature learning, and channel selection can contribute to classification performance.

Our proposed EEG-GraphNet project builds on this research by focusing on graph construction, interpretable EEG features, cross-subject and cross-session performance, and fair comparisons with established models. The aim is to investigate whether functional channel connectivity can support reliable and efficient four-class motor-imagery classification on BCI IV-2a.

References

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